

THE KENYA NATIONAL EXAMINATION COUNCIL

KCPE 2011

250381

MATHEMATICS

TIME :2 hour

READ THESE INSTRUCTION CAREFULLY

1. You have been given this question booklet and separate answer sheet. The question booklet contains 50 question
2. Do any necessary rough work in this booklet.
3. When you have chosen your answer, mark it on the **ANSWER SHEET**, not in this question booklet.

HOW TO USE THE ANSWER SHEET

4. Use only an ordinary pencil.
5. Make sure that you have written on the answer sheet:

YOUR INDEX NUMBER**YOUR NAME****NAME OF YOUR SCHOOL**

6. By drawing a dark line inside the correct numbered boxes mark your full Index Number (i.e. School Code Number and three figure Candidate's Number) in the grid near the top of your answer sheet.
7. Do not marks outside the boxes.
For each questions 1-50 four answers are given. The answer are lettered A,B,C and D. In each case only ONE of the four answers is correct. Choose the correct answer.

10. On the answer sheet the correct answer is to be shown by drawing a dark line inside the box in which the latter have chosen is written.

Example**In the question booklet:**11.What is the value of $6(24 - 18) + 6 \times 4$?

_____6

- A. 30
- B. 25
- C. 10
- D. 28

1 [A] [B] [C] [D] 11 [A] [B] [C] [D] 21 [A] [B] [C] [D] 31 [A] [B] [C] [D]

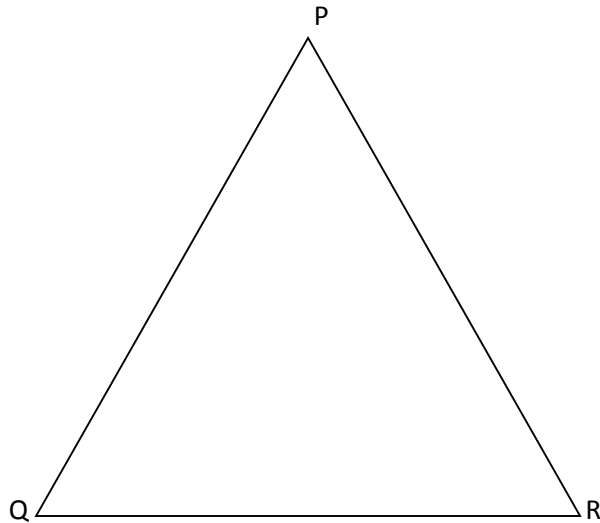
In the set,the box with letter D printed in it is marked.

11. Your dark line **MUST** be within the box.
12. For each question ONLY ONE box is to be marked in each set of four boxes.

The Kenya National Examinations Council, 2011.

1. What is 9301854 written in words?
 - A. Nine million three thousand and one, eight hundred and fifty four
 - B. Ninety three and one thousand, eight hundred and fifty four
 - C. Nine million three hundred and one thousand eight hundred and fifty four
 - D. Nine hundred and thirty thousand eighteen hundred and fifty four
2. What is the value of $\frac{2(6^2 + 4^2)}{2(7 - 5)} - 8$?
 - A. 2
 - B. 14
 - C. 18
 - D. 24
3. What is 4.59954 written correct to three decimal places?
 - A. 4.599
 - B. 4.6
 - C. 4.60
 - D. 4.600
4. What is the L.C.M. of 30,45, and 60?
 - A. 15
 - B. 135
 - C. 180
 - D. 540
5. What is the place value of digit 4 multiplied by the total value of digit 3 in the number 57438?
 - A. Ones
 - B. Tens
 - C. Hundreds
 - D. Thousands
6. Jabet bought the following items:
 - 3 packs of maize flour at sh 90 each
 - 2kg of beans for sh 170
 - 1 $\frac{1}{2}$ kg of potatoes sh 40 per kg
 - 2 loaves of bread at sh 34 each
 If she had sh 800, how much money was she left with?
 - A. Sh 62
 - B. Sh 232
 - C. Sh 466
 - D. Sh 568
7. What is the value of x in the equation $\frac{2(x + 1)}{3} - 4 = 6$?
 - A. 14
 - B. 10
 - C. 8

- D. 4
8. The area of a square is 3844cm^2 . What is the length of each side of the square?
- 1922cm
 - 961cm
 - 67cm
 - 62cm
9. Which is the correct order of writing the fractions $\frac{2}{5}$, $\frac{4}{15}$, $\frac{1}{6}$, $\frac{1}{2}$, $\frac{2}{3}$ starting from the smallest to the largest?
- $\frac{4}{15}$, $\frac{2}{5}$, $\frac{2}{3}$, $\frac{1}{6}$, $\frac{1}{2}$
 - $\frac{2}{3}$, $\frac{1}{2}$, $\frac{2}{5}$, $\frac{4}{15}$, $\frac{1}{6}$
 - $\frac{1}{2}$, $\frac{2}{3}$, $\frac{2}{5}$, $\frac{1}{6}$, $\frac{4}{5}$
 - $\frac{1}{6}$, $\frac{4}{5}$, $\frac{2}{5}$, $\frac{1}{2}$, $\frac{2}{3}$
10. In the triangle PQR below, construct a bisector of angle PQR to cut line PR at M. The bisector of angle QPR to cut line PR at N. The two bisectors intersect at point X. Join RX.



What is the size of angle RXM?

- 58°
 - 60°
 - 65°
 - 117°
11. How many fenced posts, spaced 5m apart are required to fence a rectangular plot measuring 745m by 230m?
- 391
 - 390
 - 195
 - 196

12. Awinja bought a pair of shoes for sh 810 after getting a discount of 10%. What was the marked price of the pair of shoes?

A. Sh 81
 B. Sh 729
 C. Sh 891
 D. Sh 900

13. The table below shows the amount of milk delivered by a farmer to the dairy in 6 days

Days	MON	TUE	WED	THUR	FRI	SAT
MILK IN LITRES	20	17	21	15	21	21

What was the median sale of milk, in litres for 6 days?

A. 18
 B. $19 \frac{1}{3}$
 C. $20 \frac{1}{2}$
 D. 21

14. Mutiso and Oluoch shared the profit of their business such that Mutiso got $\frac{3}{5}$ of the profit.

What was the ratio of Mutiso's share to Oluoch's share?

A. 3 : 2
 B. 5 : 3
 C. 3 : 5
 D. 2 : 3

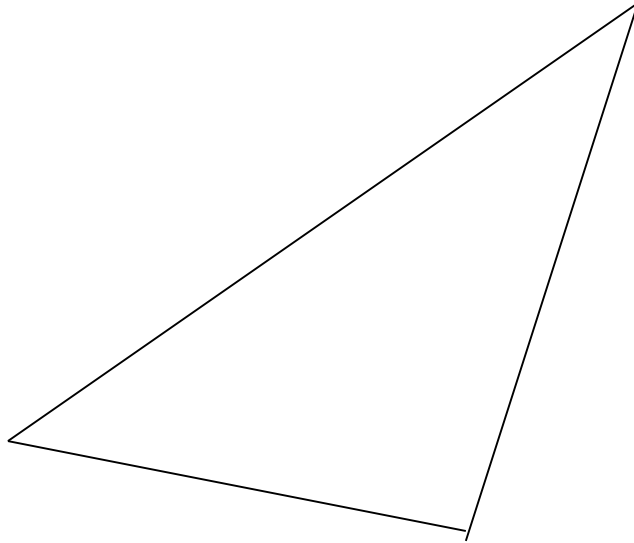
15. What is the value of $\frac{0.5 + 0.2 \div 0.25}{0.2}$?

A. 14
 B. 6.5
 C. 4.5
 D. 2.75

16. Mulwa had 5 one thousand shillings notes, 7 five hundred shillings notes, 10 two hundred shillings notes and 6 one hundred shillings notes. He then changed the money into fifty shillings notes. How notes altogether did he get?

A. 555 000
 B. 11 100
 C. 2 220
 D. 222

17. The figure below is a map of a village drawn to the scale 1 : 25 000



What is the perimeter of the village in kilometers

- A. 6 000
- B. 600
- C. 60
- D. 6

18. A cylindrical container has a circumference of 176cm and a height of 40cm. What is the volume of the container in cm^3 ?

(Take $\pi = 22/7$)

- A. 394 240
- B. 98 560
- C. 7040
- D. 3520

19. What is $\frac{1}{2} (3x + 4y) + \frac{1}{5} (2x + 7y) - 1\frac{1}{4}x - \frac{1}{2}y$ in simplified form?

- A. $\frac{13}{20}x + 2\frac{9}{10}y$
- B. $\frac{13}{20}x + 10\frac{1}{2}y$
- C. $3\frac{3}{20}x + 3\frac{9}{10}y$
- D. $4\frac{1}{4}x + 2\frac{9}{10}y$

20. The figure below is a sketch of a triangle XYZ in which angle ZXY = 50° , Angle YZX = 70° and line ZX = 6cm.

Which of the statements below leads to the correct construction of the triangle?

- A. Use a ruler to draw line ZX = 6cm long and drop a perpendicular from Y to ZX.
- B. Use a ruler to draw line ZX = 6cm long a pair of compasses to construct angle ZXY = 50° and YZX = 70°
- C. Measure and draw the angles ZXY = 50° and YZX = 50° Using a protector and draw line ZX = 6cm long
- D. Use a ruler to draw line ZX = 6cm long. Use a protector to mark of the angle of 50° at X Let the lines formed by the angles meet Y.

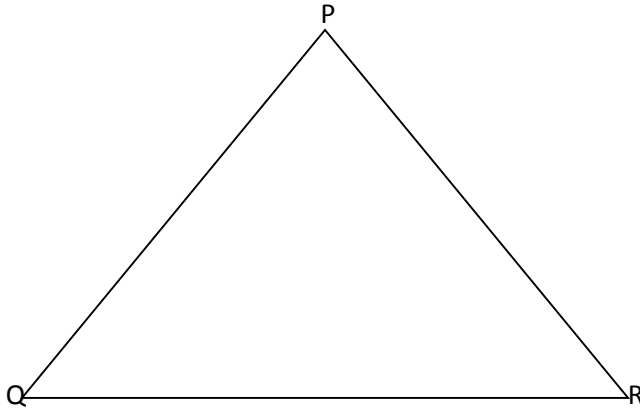
21. Each of the diagonals of a rectangular flower garden is 65m. If one side of the garden measures 25m, what is the measurement of the other side?

- A. 90m
- B. 60m
- C. 40m
- D. 20

22. A meeting was attended by 150 people. Out of these, 0.4 were men, 0.2 were children. How many more children than women were there?

- A. 69
- B. 78
- C. 99
- D. 129

23. The triangle PQR shown below has been drawn accurately.



What is the size of angle QPR

- A. 95°
- B. 85°
- C. 50°
- D. 45°

24. Mwarua is paid Sh 3 750 after working for 25 days. How much money would he be paid if he does not work for four days?

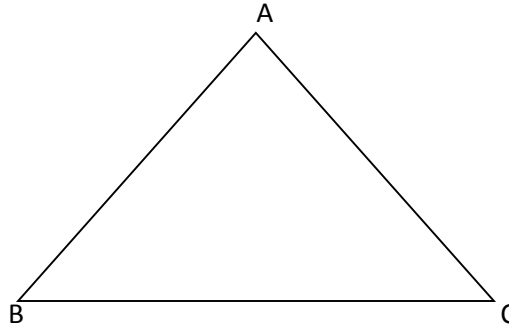
- A. Sh 600
- B. Sh 4 464
- C. 4 350
- D. 3 150

25. A family uses 5 decilitres of milk each day. How many litres of milk altogether would the family use in the months of June and July?

- A. 305
- B. 30.5

- C. 30.0
- D. 3.05

26. In the triangle ABC below, construct perpendicular from A to meet line BC at N



- A. Line AN bisect line BC
- B. Angle BAN is equal to line BC
- C. Angle ANB is equal to angle ANC
- D. Line AB is equal to line BN

27. The graph below shows the journey made y a social worker on certain day

Between which two places was his speed the highest?

- A. Home and school
- B. School and health center
- C. Health center and the market
- D. Market and Home

28. What is the value of $\frac{p(2r+q)-r}{q}$

$p=3$, $q=4$ and $r=p+q$?

- A. $8\frac{5}{7}$
- B. $6\frac{4}{7}$
- C. $2\frac{2}{7}$
- D. $\frac{2}{7}$

29. What is the surface area of a cylindrical rod of height 17cm and diameter 14cm?
(Take $\frac{22}{7}$)

- A. 748cm^2
- B. 902cm^2
- C. 1056cm^2
- D. 2728cm^2

30. What is the value of $2\frac{1}{2} \div \frac{2}{3} + \frac{5}{7} - 1\frac{2}{5}$ of $\frac{5}{6}$

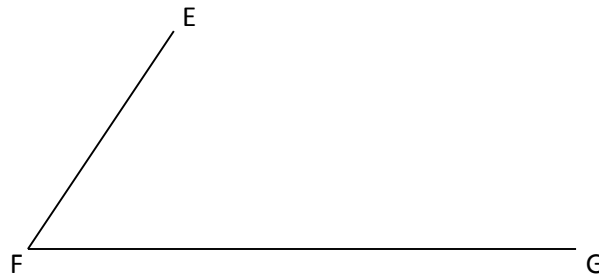
- A. $3\frac{5}{24}$

- B. $2\frac{23}{168}$
- C. $1\frac{97}{336}$
- D. $1\frac{1}{8}$

31. Halima bought 50 bananas @ sh 3 each. She spent sh 75 for transportation. During transportation 5 bananas got spoilt but she sold the rest making a 20% profit. For how much did she sell each banana?

- A. Sh 4.00
- B. Sh 5.40
- C. Sh 5.60
- D. Sh 6.00

32. Two sides of a parallelogram EFGH have been drawn below. Complete the parallelogram EFGH. Draw diagonals EG and FH to intersect at J.



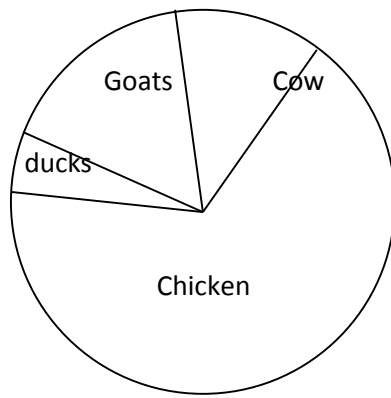
What is the length of line FJ?

- A. 2.7 cm
- B. 3.5cm
- C. 4.4cm
- D. 6.5cm

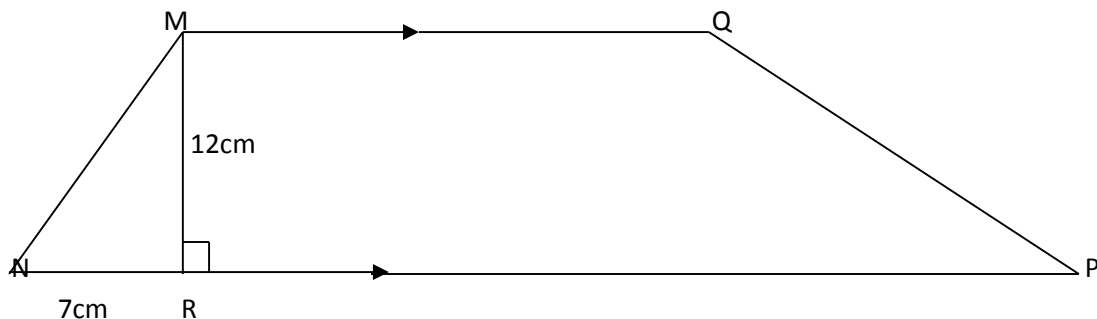
33. Mutuma left Mombasa on Tuesday at 6.30p.m. and took 8 hours 45 minutes to reach his home. On what day and at what time in a 24hour system did he reach home?

- A. Wednesday 0315h
- B. Wednesday 1515h
- C. Tuesday 1515h
- D. Tuesday 0315h

34. The pie chart below represents the population of 1800 animals in farm.



35. A car travelled 216 km at an average speed of 48km/h. On the return journey the average speed increased to 27km/h. Calculate the average speed, in km/h, for the whole journey?
- 57.6
 - 60
 - 28.8
 - 68.6
36. Which one of the statement below is a property of a right angled triangle?
- All sides are equal
 - Adjacent angle are supplementary
 - Two of its sides are perpendicular
 - The longest sides of the triangle is opposite the smallest angle
37. A mathematics text book has 97 sheets of paper and a cover. Each sheet of paper has a mass of 4 grams and the cover has a mass of 20g. Find the mass of the book in kilograms.
- 0.408
 - 4.08
 - 40.08
 - 408
38. The diagram below is a trapezium MNPQ. Line MQ is a parallel to line NP. The length of the line MQ =8cm and that of line NR =7cm. The perpendicular line MR =12cm.



If the area of the trapezium is 198cm^2 what is the length of RP?

- A. 15cm
- B. 18cm
- C. 25cm
- D. 32cm

39. Ali is now two years older than Martha. If Martha's age is represented by x , what will be total ages after 10 years?

- A. $2x + 22$
- B. $3x + 20$
- C. $x + 22$
- D. $2x + 18$

40. A football match was attended by 42 000 men. Then number of women who attended was 27 000 less than the number of men and 12000 more than the number of children. The entire fee for adults was sh 100 and for children was sh 50. How much money was collected altogether?

- A. Sh 11 700 000
- B. Sh 7 500 000
- C. Sh 7 050 000
- D. Sh 5 850 000

41. In the figure below EFG is a straight line. Line GH and FH are equal and line HI and FI are also equal. Angle GHF is a right angle HIF is 32° . What is the size of angle EFI?

- A. 45°
- B. 61°
- C. 74°
- D. 103°

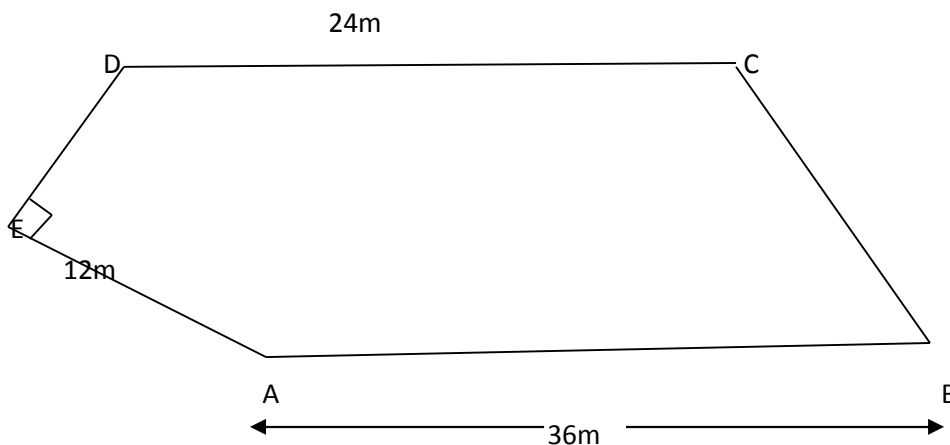
42. The table below shows part of the Tariff for Ordinary Money Order and Posta pay.

Range	Ordinary	
Money order	Postapay	
	(sh)	(sh)
7501 -100000	250	550
100001 -200000	300	675
20001-35000	400	800
35001 -50000	450	1200
5000 – 75000	550	1600

Karimi has two children in one school. To pay for their school fees he sent sh 8900 by Ordinary Money order and sh 15400 by Postpay. How much money would he have saved had he bought one Ordinary Money Order to pay for all the fees?

- A. Sh. 125
- B. Sh. 400
- C. Sh. 525
- D. Sh. 925

43. Nina is paid a basic salary of sh 8000 as a sales agent. In addition she is paid a 5% commission for goods sold above sh 15000. In one month she eared sh 12000 altogether. What was the total sales?
- Sh 255000
 - Sh 95000
 - Sh 80000
 - Sh 65000
44. What is the next number in the pattern 4, 9, 25, 49, 121, 169, _____?
- 289
 - 256
 - 225
 - 196
45. The marked price of a motorcycle was sh 3 0000 but a discount of 5% was allowed for cash payment. Taabu bought the motorcycle on hire purchase terms by paying a deposit of sh 8500 followed by ten equal monthly instalments of sh 2400 each. How much money would taabu have saved had she bought it for
- Sh 4000
 - Sh 2500
 - Sh 1500
 - Sh 28 500
46. Figure ABCDE below represents a vergatable garden in which $AE = 12\text{m}$, $AB = 36\text{m}$ and $CD = 24\text{m}$. Angle DEA is a right angle. The distance from A to D is 15m. A perpendicular distance from C to AB is 10m.



What is the area of the garden?

- 474m^2

- B. 390m^2
- C. 354m^2
- D. 300m^2

47. A tailor made 48 pieces of uniform. Half of the number of the uniforms was each made using $1\frac{1}{4}$ metres of materials. A quarter of the remainder was each made using $1\frac{1}{2}$ metres of materials and the rest was each made using $1\frac{3}{4}$ metres of the materials. The tailor also fixed a logo made using $\frac{1}{16}$ metres of the material on each uniform. How many metres of materials did the tailor use?

- A. $73\frac{1}{2}$ metres
- B. 72 metres
- C. $70\frac{9}{16}$ metres
- D. $70\frac{1}{2}$ metres

48. The table below shows the train fares for Nairobi – Mombasa route.

- A. Sh 119560
- B. Sh 151300
- C. Sh 156100
- D. Sh 164080

49. A man deposited sh 50000 in a bank for 2 years. The bank paid compound interest at rate of 10% per annum. How much money was in his account at the end of two years?

- A. Sh 10500
- B. Sh 55500
- C. Sh 60000
- D. Sh 60500

50. The figure below represents a pattern

Which one of the following is the next shape in the pattern?

I